

How can HCI support end-of-life care? Critical perspectives on sociotechnical imaginaries for palliative care

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ABSTRACT

Designing technologies in the palliative care environment is difficult due to the sensitivity of the context and vulnerability of the patients and their families in a critical stage of life. We conducted two focus groups with 9 healthcare workers to capture sociotechnical imaginaries to support physical, psychological, social and spiritual aspects of the patient care in the palliative environment. The findings reveal recurring themes of care: expressing respect in safe spaces, transitioning into the unknown and experiencing home again, taking opportunities for uplifting care, communication for empowerment and risk mitigation, and connection to support networks. These are discussed through a lens of the materiality of care provided to patients, where tangible materials of care are entwined with patients' values and culture. This in turn offers insight into opportunities for palliative care technology.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI.

KEYWORDS

palliative care, virtual reality, end-of-life, sociotechnical infrastructure, materiality of care, ethic of care

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1 INTRODUCTION

Designing for end-of-life brings up many questions and warrants research in Human-Computer Interaction (HCI). Challenges for HCI emerge from clinical contexts such as specialist palliative care (PC), ethics of technology interventions in such a critical stage of life, and ethics of participatory research with vulnerable individuals, including people with palliative care needs and their families. Yet, there is potential to improve the delivery of palliative care through technologies that are increasingly shaping people's experiences at home and in healthcare. So, *in what ways may technology play a role in palliative care, if any?* We address this challenge in our research through engagement with PC healthcare professionals, as part of larger, multi-perspective participatory design research project.

The World Health Organization defines PC as “an approach that improves the quality of life of patients and their families facing the problems associated with life-threatening illness. It prevents and

relieves suffering through the early identification, correct assessment and treatment of pain and other problems, whether physical, psychological, social and spiritual” [25]. The COVID-19 crisis provided stark evidence that technologies can be instrumental in the context of care [21], for example when remote care is required to secure the wellbeing of those who are at higher risk of infection, when PC patients face mobility challenges due to lack of energy or other health problems, and for strengthening connection with loved ones. These factors reinforce the *Australian 2018 National Palliative Care Strategy* that stresses the need to harness personalized, technology-enhanced interactive experiences to improve PC outcomes [23].

Our project started with an interest in immersive technologies, like Virtual Reality (VR), for supporting palliative care to assist in physical (e.g. pain), psychological (e.g. anxiety), social (e.g. relatedness) and spiritual (e.g. faith-based care) aspects of care for patients with a life-limiting diagnosis. This was based on recent evidence suggesting the potential [15] and feasibility [5] of VR for alleviating pain for cancer and PC patients with complex diagnosis. However, we note that VR or other head-mounted devices (HMD) might not be comfortable for patients with severe physical or cognitive conditions, for example due to the weight of the HMD for frail patients, HMD not being practical to use for patients with a severe head or hand tremor, or for patients with cognitive decline. Additionally, it may not be feasible for VR to facilitate all the various dimensions of PC. Therefore, we broadened the scope of our work to consider a PC technology infrastructure situated in the social context of care, involving patient’s family, community, healthcare staff and volunteers. By infrastructure, we broadly refer to physical, social, and organisational systems and resources that support the provision of care [3]. We include in this definition the routines and practices embedded in the built environment, technologies, transportation, social and other relevant services that shape the practice of “care” relative to the goals and conditions that are meaningful in a context, for example PC. The technology components of the infrastructure in our case may include a range of patient-facing devices such as VR, tablet, or mobile apps to assist PC patients based on their needs. We address the following research question:

RQ What are the experiential aspects of palliative care that may be enhanced through technology infrastructure, particularly through VR?

To answer the question, we conducted two focus groups as part of a larger participatory design research project. Nine healthcare staff currently working in a PC facility participated in focus groups to discuss their practices, philosophy and experiences of care for PC patients. They were also encouraged to speculate about possible technology imaginaries such as VR, which might fit well with their existing practice. The focus groups identified 5 key themes: (1) outlining the way practitioners embed and express respect in safe spaces for patients, (2) helping patients transition into unknown and experience home again, (3) taking opportunities for uplifting care, (4) communicating for empowerment and risk mitigation and (5) connecting to a support network. The findings contribute an empirical account of palliative care from the perspective of clinical and allied services practitioners, such that it could guide HCI efforts

to consider ways of materialising care. We also contribute a future vision of PC technologies.

2 RELATED WORK

2.1 Essential Palliative Care (PC) Principles

PC is often described as a medical specialty in the context of terminal illness and the transition to the end of life [16], and is positioned in the space between cure and acceptance of death [7, 16]. PC is, above all, person-centred, meaning that it is structured around the needs and preferences of each individual patient and their carers, enabling them to take as much control as possible [30]. As such, PC aims to empower patients to reclaim their care in the final stage of life. Therefore, PC includes, but is not limited to, a chronic disease framework in which care is viewed as a collaborative endeavour involving patients, families and health professionals working in partnership, with an emphasis on supporting self-management. Along this line, Alex Broom critiqued PC from the sociological standpoint, asserting that PC should not be viewed as “help” or “assistance” but rather a vital connection, necessitating a view beyond medicalisation of suffering [6]. To achieve that, we must better understand how PC is currently shaped and what constitutes the PC experience cultivated by the healthcare staff in the PC context.

In Australia, a core tenet of PC is that patients and carers should be involved in shared decision-making and planning for current and future care [24]. This requires that patients and families are sufficiently informed about the benefits and risks associated with each management option and have their needs and preferences understood and documented by the healthcare team. The role of technology outlined by the National Palliative Care Strategy in Australia [23] specifically calls for designs and services to support PC at home for longer through the use of home-based services. Given that PC is underlined by a holistic perspective towards physical, psychological, social and spiritual care, any approach to technology design must endeavour to understand how such vision is contextualized and embodied in the PC journey. We next review some existing research at the intersection of PC and HCI, and specifically the relevance of immersive technologies such as VR.

2.2 Palliative care, HCI and immersive technologies

HCI research in relation to PC often falls into one of the three categories: managing care delivery [19, 27]; healthcare staff training [18, 32]; and design & assessment of patient-centered technology interventions, such as robots [20]. However, there is a clear need to elucidate a holistic understanding of HCI’s role in PC to provide temporal and embodied experience with various psychological, spiritual and social dimensions. This experience is shaped by everyday interactions in a care ecosystem involving healthcare professionals, family and loved ones, and other affiliates (such as PC volunteers) who all play a role in care giving. Attempting to consider whether or how a technology infrastructure may play a role in PC is incomplete without addressing this gap in HCI.

With regard to patient-centered technologies for PC, there has been an interest in using virtual reality (VR) and immersive experiences [9, 13, 17]. Immersive technologies like VR are increasingly explored in relation to acute and chronic pain management [2]

for adults [15] including older adults [14], children [1] and more recently in the context of palliative care given that alleviating physical pain is an important aspect of PC [5]. Additionally, VR has been used for rehabilitation and physiotherapy [12]. Through cognitive, affective and sensory inputs, VR enables a sense of presence that is advantageous to self-regulating actions and strategies. For example, Google Earth in VR has been used to enable terminal cancer patients to view immersive videos of “memorable places.” This intervention aims to alleviate pain through distraction, help manage tiredness and anxiety, among other symptoms [22]. Distraction here is considered a passive strategy to divert the patient’s attention from their symptoms to an alternative stimulus, for example, media or interactive features in VR [2, 4]. An active strategy may utilize sensors and biometric input in interactive VR, whereby the user provides input such as heart or respiration rate to trigger changes in the virtual environment. This approach has been explored for helping patients learn skills for long-term managing of symptoms such as anxiety and pain [29]. Finally, VR has been explored extensively in hospitals and aged care facilities for enhancing the experiences of individuals through enriching virtual [33] and media experiences [34].

Overall, the evidence points to the feasibility of VR for PC symptom management. It also identifies a key role for formal caregivers, such as healthcare staff, to identify, deliver and assess the care experience, as well as mitigating risks of introducing technology into the care ecosystem. Through our research we aim to explore the roles future VR-mediated experience might play in the PC context, while building an understanding of the role of HCI in supporting patients manage their pain, psychological, social and spiritual (PPSS) experience.

3 METHOD

We conducted two focus groups with staff members from a Palliative Care unit in Sydney, Australia with a total of nine participants, with each discussion facilitated by two HCI researchers. The two sessions were scheduled around the availability of staff and conducted remotely to comply with infection control and COVID-19 safe protocols at the time of study in 2021. As seen in Table 1, focus group participants ranged from those with clinical experience such as PC nurse practitioner to specialist professionals such as dietitian and bereavement counsellor. This ensured participants contributed diverse insights into the research. The number of years participants worked in palliative care ranged from 5 to over 15 years, in multiple institutions. Therefore the findings, while situated in one current institution, provide a holistic picture of PC in the Australian context. Participants were recruited through advertisement at the clinic and all gave written informed consent. The study was approved by the ethics committee at HammondCare Governance Office, ref 2021/ETH10958.

We discussed how the staff members support adult patients in managing their PPSS experiences. Participants described typical strategies used with PC patients, particularly new patients, barriers to implementing these strategies for both patients and caretakers, patient loneliness and the role of family, friends, community nurses, volunteers, or others important in the patient’s care. Next, we explored opportunities for creating meaningful and personalized platforms for mediating PPSS, such that it is adaptable to individual

patient values, assets and needs. Finally, participants were invited to speculate technology directions if they had a “magic wand” to help their patients. This encouraged them to think freely, disregard any technical or other limitations and imagine possibilities to improve PPSS experiences.

The focus groups were recorded and transcribed. The transcription was anonymized by assigning a code to each participant and all identifying details of staff members were removed to ensure participant privacy. Three researchers from outside of the PC institution stored data securely and conducted a reflexive thematic analysis (RTA) by following the six steps suggested by Clarke and Braun [8], including dataset familiarisation, data coding and initial theme generation by one researcher, theme development, then refining and defining by all three researchers, and finally write up. For example, creating safe spaces to build trust with patients, was coded “staff creating a safe space”, clinical care beyond treating physical pain symptoms was coded “holistic experience” and capturing what is meaningful to the unique patient was coded “understanding individual needs”. These formed a larger theme, “Holistic care for the individual.” Overall, we defined five themes to illustrate how staff demonstrate care for PC patients. We then classified potential opportunities identified by the participants (using a magic wand) into these themes.

4 FINDINGS

4.1 Expressing respect in safe spaces

Participants discussed a holistic care approach that initially focuses on physical symptoms, then shifts to understanding other needs of the patient. Part of this holistic approach entails creating a safe environment to build trust, so that patients can share their needs and values. This care delivery progresses without necessitating a specific goal or timeframe, *“So they don’t feel like on this ward that it’s a production line, and we’ve got to get people out as fast as possible... that whole environment allows people to feel safe”* [P8]. One participant noted trust is bridged through small acts demonstrating value and respect by making the mundane special: *“We have some beautiful fine bone china here and sometimes I take someone a fresh cup of coffee or tea in this amazing fine bone china, it both brings them joy and relief, and they feel special. And that small little act can be a bridge builder for developing a relationship, where they can talk about things that are hard and difficult.”* [P8]. Several approaches for softening emotional boundaries are used to invite conversation including art therapy, *“When they’re sitting around a table doing a painting or something, that’s where they feel very safe to talk about things that really matter, things that are really hard. So that is another very powerful non-pharmacological method.”* [P8]. Participants noted that creating a safe space also requires interpreting non-verbal cues and a sensitivity to the patient’s boundaries, *“that’s often a lot about timing or... our body language, our listening to them to hear what is appropriate for them... and not to keep pushing them in a conversation... but to respect that and keep checking in that it’s okay to keep talking to them”* [P8]. The purpose of creating safe spaces is to initiate challenging conversations to understand what would be meaningful for patients at this point in their life. *“Finding out what’s important to you... where do you feel safe? And then how*

Table 1: Role or job title of focus group participants

Focus Group 1	Focus Group 2
P1 Senior Staff Specialist	P6 Clinical Nurse Educator
P2 Dietitian	P7 Social Worker, Bereavement Counselor
P3 Osteopath, Researcher	P8 Nurse
P4 Clinical Nurse Consultant	P9 Nurse, Director Operations
P5 Palliative Care Nurse, Researcher	

can we facilitate that through what we currently have on the ward” [P9].

In the magic wand activity, participants speculated about immersive experiences for reminiscence, using personalized libraries “in an environment [such as] camping trips... back in that space and take them away from the reality of lying in a bed because a lot of our patients physically can’t move either.” [P9]. The virtual library fulfills the inclinations of the individual “A suite or a library of different types of, VR application... For example, if spiritual distress was something that was important to the individual, or different types of games, and you could almost have genres within this library of where people would like to go and what would most suit them” [P3].

4.2 Transitioning into unknown and experiencing home again

Transitioning into the unknown can add to the psychological distress of new patients, “Patients come to us with a high level of anxiety which, you know, is linked with pain, emotional pain... over time, they start to feel safe here. And we see that anxiety reduced” [P7]. The participants spoke about how existing technologies may ease the change in the patient’s lives, “Many patients have discharged to nursing homes and that can be very stressful for patients and families. And I guess it’s the unknown that is very scary. So it has been good to be able to do some sort of virtual tours of the homes on [the computer] and show them... the nursing home.” [P7]. Technology may allow patients who wish to return home a chance to return and reconnect with positive memories, “Imagine living in a bed and knowing that you’re going to die imminently or soon so... [for] a lot of them, the home is in another country; to be able to take them back to that other country would be wonderful” [P9]. Possible virtual home tour was discussed, “you [can] have a program where a family member walks around and [films, so the patient] can walk around virtually in their home for the last time, because there’s 40 steps to the front door [but] that’s not realistic for them” [P2]. The participant reflected on not only feelings of home through recreation of the physical environment, but also special moments of gathering in the home, “If you could feel like you’re at home having Christmas, [especially] with all the lockdowns in the last two years... if you can sit down at dinner table for one last time... with your whole family... that would be amazing... so you can sit down and have a conversation with your grandkids when [they are] overseas... That would be really nice in a more virtual reality way as opposed to Teams or Skype” [P2].

Participants described home as a place where memories are collected and rituals are performed, such as coming together through celebration. They imagined technology may help recreate rituals in a meaningful way and fulfil a longing to be in a familiar place enabling them to look back on their lives. Participants spoke about

volunteers helping create biographies, which also give a path to prepare patients to reflect through their legacy, “[volunteers] have been trained to do the biographies... they’ve done a lot of [biographies] via Teams or Skype... that’s been really beneficial [during COVID].” [P5]. All of this is done for comfort.

4.3 Taking opportunities for uplifting care

PC patients may be restricted to their rooms or beds. Changing their environment can be uplifting. Participants engage with the patients’ individual needs or interests: “During COVID, there was a lady that couldn’t go outside. She likes these red leaved trees. So we got on the computer and printed up all these pictures of... red leaved trees [that we] stuck on her wall, and she loved just looking around the room at that rather than a blank wall.” [P6]. These opportunities were investigated and identified through personal interactions: “He was coming to end of life care and was extremely in pain... [We] slowly built up his trust... and then one day he said he had loved to go outside [but] at that point he still couldn’t. So, we put his bed outside into the car park, but he had a view of this amazing tree, he felt the wind on his face... Something was set off in his head... he then started to want to have a shower, get up out of bed, get dressed, and he eventually went to an aged care facility” [P9]. Volunteers also play an important role in opening possibilities of care, “They’re not asking people about their pain or their bowels or their whatever, [volunteers are] just there to chat about the weather or whatever [the patients] want.” [P5]. These volunteers and staff also enable patient wish fulfillment through the Dreams Project, an initiative with the clinic. For example, one patient who had a dream to go to Balmoral Beach... and the team took the patient and his wife, so they could share a lunch together: “So he just had a lovely time out just for a few hours, and it was very uplifting for him.” [P7]. The participants reflected how this initiative helped redefine the palliative experience, “In the Dreams Project, it just disrupts the illness journey... reframes or changes those pain pathways... and that’s really the quite a great thing to be able to do, not needing a medication for that.” [P9]. Using their magic wand responses, the participants wanted to give their patients sensory experiences based on their own desires, “If they loved flying or racing a car... do that from sitting in a bed, [or experience] the smell of home cooking or chocolate chip cookies” [P6] or go to a sensory experience room that takes patients into “a sacred space [away] from the hustle and bustle of the clinical area” [P9].

4.4 Communication for empowerment and risk mitigation

The means of communicating accessible information are key to addressing health literacy, which refers to both how people understand information and apply it to their lives. Health literacy is also

linked to how symptoms are managed, for example “*understanding that eating doesn’t [always] give you energy [when in PC]... accepting quality of life over certain other things... it can be a barrier sometimes to giving best care or best strategies to manage symptoms*” [P2]. The competing sources of health information can play into patients and carers assumptions, “*Some cultures hold [assumptions], particularly about opioids... because of [the patient’s] experience or what they’ve seen on the media*” [P5]. Barriers to communicating complex medical information may result in patients and carers searching online for information, which amplifies the risk of misinformation. Participants reflected on how they provided information amongst multiple sources of information, “*If they have a well-meaning relative, [who] comes in and knows how to Google or read it on Facebook. Participants want to “redirect them to what would be evidence-based advice”* [P5]. Participants suggested a curated source of reliable information to empower patients and their carers to mitigate the risk of misinformation.

The staff suggested incorporating trusted information into an education platform that is personalized to the individual, so as to empower them. “*They can put in their symptoms, their questions of distress, and it actually takes them through to an area that will give them information on that or... say, well, it’s really normal for people not to eat so much [in this condition], it’s more about quality rather than quantity*” [P5]. Such a technology platform should be accessible and easy to use by people receiving PC, “*Going to Google is something that might be really foreign to a lot of older patients. And they just might be sitting there waiting to get book, or they need a physical pamphlet telling them that information. So, I think probably a combination of having a reputable, trusted source, like... an app... where you can go and access all the [palliative care] information that you know is trusted*” [P2].

4.5 Connecting to a support network

Connecting to friends and family is balanced with the energy of PC patients, for whom even talking can be exhausting. Texting helps to control the conversation depending on the pain and energy level of the patient, “*They find it really helpful to text their loved ones because [of] the effort [needed for] talking... For the patients to be able to contain those conversations, keeping contact with a lot of people and feel like... they’ve got a bit of control over how they’re doing that rather than just answering the phone*” [P7]. Furthermore, the presence of a loved one and listening to a conversation between others also provides feelings of relatedness, “*having their support networks around them is crucial to that person’s feeling safe and enjoying their life... just being in a room with your family members, not be talking.*” [P9]. Video calls opened possibilities of communication during COVID-19 social distancing, and were made feasible through the assistance of volunteers, “*With the help of volunteers, we were able to organize some video visiting... people could have something to look forward to... We set up the appointment... [and] the volunteer who was doing that, they often went above and beyond when they came in to do that for the patients. And that was significant as well.*” [P7]. While video call appointments give a predictable means to connect, the issue of feeling related and present remains salient, “*The depth and the senses [are missing], because you can’t smell [the person,*

place.] Especially if it’s in a different country... But it’s like a visit from their home” [P9].

One participant said the magic could give the feeling of family members in the same room through a simple-to-wear device, “*almost like I just need to wear my glasses, and then I feel like I’m in this room already with you guys. I think I’ll be really happy that I can see my family members next to me. And it’s following me... like the hologram style, but I’ve got it on my eye-wear only, not the full VR headset... I feel like it would bring some connectedness to people somehow some way that feels real rather than on screen.*” [P4].

5 DISCUSSION

Findings from our study revealed that any technology infrastructure, including VR and other immersive technologies, must be designed with an understanding of the holistic approach of PC, including physical, psychological, social and spiritual (PPSS) care. This is an important contribution of our work that is under-explored in the literature, particularly in HCI. The experiences of staff as critical caregivers is instrumental in grounding this understanding and developing an ethical vision of technology infrastructure. We had initially imagined a project to develop a VR platform consisting of multiple experiences to address PPSS needs, but our findings suggest that apart from the intervention form, we must integrate experiential qualities and personalisation into technology mediated PC in a holistic way. Consequently, two points emerge, which we discuss in this section. The first is the materiality of the care that is provided, and its ability to reconnect the caregiver to the personhood of the patient. The second is a vision of the future of technology infrastructure in PC that recreates subtle qualities of experiences that can have an uplifting effect on the patient. A recent review of PC during COVID-19 highlights that the use of new technologies as modes of support must be tailored to the needs of individuals and also recognize difference and diversity across patients and families [17]. This is particularly challenging in the context of VR, as previous studies required considerable resources to tailor a meaningful VR experience for individual patients, and so are not scalable [35]. Personalisation must be balanced with building a patient’s sense of control which can be therapeutic in its own right, and fits better with a longer-term and/or home-based care [10, 11].

5.1 Materiality of care

Through the experiences described by the participants, we can identify a material quality to care arising from the tangibility of experiences. The care-full interactions are facilitated through the materials that are close by, and the meaning is ascribed to them by patients through experience and culture: a combination that creates an entangled whole [36]. This can be seen through the examples shared in our focus group: as caregivers change the rooms where a patient is unable to leave, by printing pictures of the kind of trees the patient likes, they entangle the space with fond memories; by offering tea in fine bone china they make a mundane experience special.

We propose that PC technology does not need to replicate or even augment the same acts of care that are provided by staff and volunteers. Rather, it should aim to create an experience of care

that respects and enriches the materiality of existing practices, with interplay between the patient, the caregiver and the technology becoming the point of departure for design [36]. Indeed, we do not propose that a technology mediated intervention is needed in PC. Instead, we stipulate that PC technology may bring materiality into the experience of care while aligning itself with the practice of care. It is worth considering how such materiality plays into the experience of PC in the future as interactive technology becomes ubiquitous. Here it is useful to reflect on questions proposed by Joan Tronto [31] to gauge our “caring judgement” around “elements of care.” We should question our *attentiveness* to needs of the care recipient, *responsibility* and *competence* of caregivers to meet those identified needs, and *responsiveness* as the care provided alters the situation. It is critical that any decisions about materials of care be assessed against these elements to enable questions about the legitimacy of the material, its quality and contribution to ethic of care to make “daily and thoughtful judgement about caring” [31, p. 16]

5.2 Future vision of palliative care technologies

Technological interventions in future palliative care scenarios should not seek to replace the act of care that is provided by the caregiver. The *Report of the Lancet Commission on the Value of Death* outlines a major cultural change of death in the 21st Century: death had previously been an aspect of family and community settings, but in most cases it has moved to the healthcare domain [28]. Technological interventions may contribute to the further medicalisation of death; however, one aspect of PC technology should be to provide opportunities and infrastructure for the caregiver to provide relief from the circumstance, not necessarily of death (which is a part of life), but of the clinical setting.

The materials of care our participants imagined served the patient by providing subtle qualities of experiences, rather than total escape from the situation. They acted as more than a distraction, but an anchor for the patient while they are on this journey, through an affective interaction with the caregiver and the clinical setting. This is seen through the current strategies that the caregivers employ. The un-complicated experience of the patient drinking tea from a fine porcelain tea set or feeling the breeze on their face, whether they are able to travel to the beach, or just in the parking lot, provides an example of how the caregiver creates these experiences without a technological intervention.

So, how should technological interventions sit in this context? We propose that technology *has the potential* to contribute by helping the *patient as they reflect back*, making ways to find comfort, and also helping the patient *find peace and reframe their current experiences*. Technology could create opportunities for experiences that support recollection of cherished memories for reflection, e.g. virtual visit to cherished places. Providing a “home” [virtual] experience in a clinical setting is a clear opportunity, whether the “home” is in a single building, another country or even another time. An infrastructure of technological care based around this reflection will help patients to reframe themselves beyond medical symptoms.

An opportunity also exists to combine experiences with personalisation. This requires a framework of collection and categorisation of interests and desires, which could be fulfilled by drawing on

a library of experiential interventions. One clear limitation for technological intervention is the capacity of medical services to support and facilitate the experience. This could be due to limited professional or volunteer ability or availability, compared to the wide range of possible experiences that could be created. However, future affective experiences could utilize AI to meaningfully personalize technology-mediated care to provide the kind of subtle qualities of care currently provided without technology. For example, a virtual scene of a patient’s hometown, created to facilitate reflection, could be augmented with the birdsong that is relevant to the country and time of year. There are many opportunities for research to understand different elements of interactive experiences that contribute to these qualities of experience in a palliative care setting.

6 CONCLUSION AND FUTURE WORK

The participants identified technology for health literacy as an important topic, which has previously been outlined as an important issue in PC [26]. But further to this need, we found that there is an opportunity for technology infrastructure to support the patient, through immersive and interactive experiences such as VR, to find relief from the clinical experience. Our findings are situated in the Australian context, and contribute to a holistic approach as we untangle the material quality of palliative care. In doing so, we see opportunities for technologies to become part of an infrastructure of care and generate experiences of safety and familiarity when patients transition to the unknown PC environment, and to uplift care through an interconnected network of technology, services and social constructs. Previous research has already pointed to many variables which can impact sustainable implementation of VR and similar technologies in hospitals and clinics [15] such as considerations for staff training and workload, technology maintenance, access to reliable library of apps and more. The next step for this project will explore these possibilities by extending the research to PC patients, their families and carers, as well as volunteers in PC clinics. We hope to contribute to critical discussions about the role technology may or may not play in PC while ensuring the integrity and dignity of patients are preserved in a critical stage of life.

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